12) Municipal Waste Collection.

Application: public transportation

problem statement: Design routes that service all required collection points while Accounting for truck capacity road distances and depot locations

objective: Minimize total travel distance and the number of vehicles required

Waste collection routing is among the documented Applications of ACO

psuedocode

BEGIN

1. initialize:
   define depot, collection points, demands and vehicle capacity
   set parameters (ants, alpha, beta, rho, initial pheromone, Max iterations)
   initialize pheromone levels on all road segments
2. construct solutions:
   For each ant:
   while unvisited points remain:
   start a new truck route at depot
   while unvisited points fit within Remaining truck capacity:

select next point using transition probability
Move truck, add waste to load, and mark point as visited

END WHILE
return truck to depot to complete the route
END WHILE
END FOR

3. Evaluate FLEET TOURS:
calculate total travel distance for each ant's fleet
Count the total number of vehicles used by each ant

4. UPDATE PHEROMONES:
Evaporate old pheromones on all road segments
Deposit new pheromones based on solution Quality
(shorter distance & fewer trucks = more pheromone)

5. CHECK STOPPING CONDITION:
update the best global fleet route found
if max iterations reached STOP.
Else
repeat from step 2

6. OUTPUT

display best vehicles routes, Minimum distance and fleet size

END

30/10/26

output

Enter total number of waste collection points (excluding depots) : 3
Enter Maximum vehicle capacity (weight/Volume) : 5

Enter coordinates for Depot (point 0):
Depot X : 0
Depot Y : 0
Enter data for collection point 1:
point 1 X : 0
point 1 Y : 4
point 1 waste Demand : 3
Enter data for collection point 2:
point 2 X : 3
point 2 Y : 0
point 2 waste Demand : 3
Enter data for collection point 3:
point 3 X : 0
point 3 Y : 0-4
point 3 waste Demand : 2

optimization complete (50 - iterations performed)

Minimum Number of vehicles required : 2

Minimum Total distance : 20.00 units

Best vehicle Route paths

Truck 1 : Depot → point_3 → point_2 → Depot

Truck 2 : Depot → point_1 → Depot

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